



兰州大学  
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# Evolution of Lightweight Gorpcore and Design Curation for Local Brands Based on Multimodal Social Data

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## 1. Industry Landscape and Research Background

For the global fashion industry in 2026, Gorpcore is no longer just a niche subcultural symbol; it has evolved into a design philosophy that deeply influences urban lifestyles [1]. This term was first proposed by *The Cut* in 2017. Its core logic lies in integrating professional equipment originally designed for extreme outdoor environments into daily urban wear. The naming inspiration comes from “Good Old” Raisins and Peanuts,” a common energy supplement used by outdoor explorers [2]. With the acceleration of modern urban life and the increasing uncertainty of the climate environment, consumer demand for clothing has shifted from a single aesthetic orientation to the psychological assurance of “Preparedness” [4]. This shift prompted a deep paradigm fusion between outdoor functionality and urban commute aesthetics between 2025 and 2026 [1].

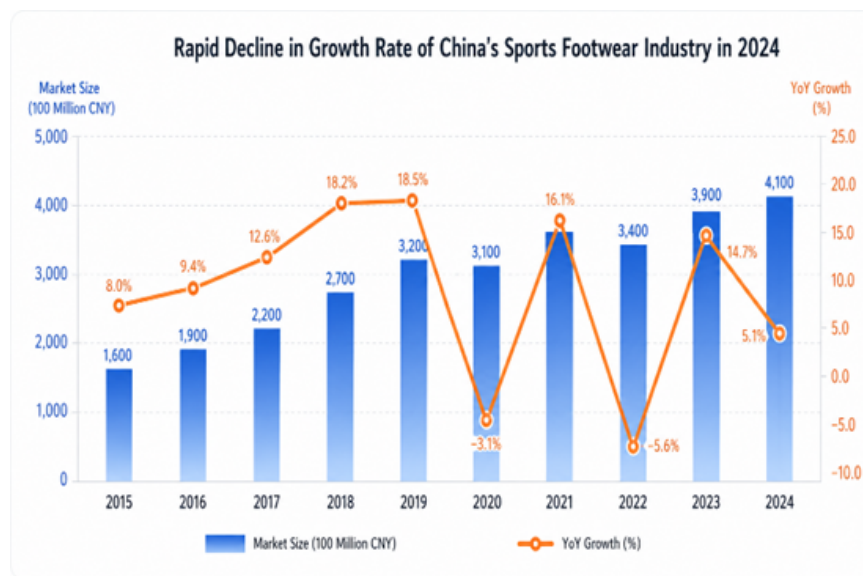


**Figure 1.** Illustration of Gorpcore style: the extension of outdoor functional apparel into urban everyday wear.

After a decade of high growth, the Chinese sporting goods market is in a critical transition period. According to the latest data from Euromonitor, the penetration rate of the Chinese sportswear market has reached approximately 50%, matching mature markets such as the US and Japan. This means the industry’s dividend period is nearing its end, and brand competition has entered the “deep water zone” of stock-based competition [6]. In this context, the Gorpcore segment has become one of the few vertical fields maintaining strong growth.



**Figure 2.** Visual reference for the lightweight Gorpcore aesthetic.



**Figure 3.** Schematic of outdoor market positioning by domestic sportswear brands.

In 2024, the size of China's outdoor sports market soared to \$55.8 billion, a 20% increase from the previous year, with sales of cycling equipment, mountain shells, and hiking boots surging by 54% and 49%, respectively [7].

For local giants like Anta and Li-Ning, Gorpcore is not just a supplement to product lines but a breakthrough point for brand premiums and internationalization strategies [8]. Anta, through its mature “single-focus, multi-brand, global” strategy, has built a rigorous outdoor brand matrix using its subsidiaries Arc'teryx, Salomon, and Kolon Sport [8]. In the 2025 fiscal year, Anta Group's revenue exceeded 80 billion RMB, with the outdoor brand group's contribution and growth rate far exceeding traditional sports categories; Descente achieved a retail volume of 10 billion RMB for the first time [9], as depicted in Figure 3.

In contrast, Li-Ning adheres to a “purist” route of “single brand, multi-category, multi-



**Figure 4.** The evolution of fashion trend analysis from expert-driven judgment to multimodal intelligent curation.

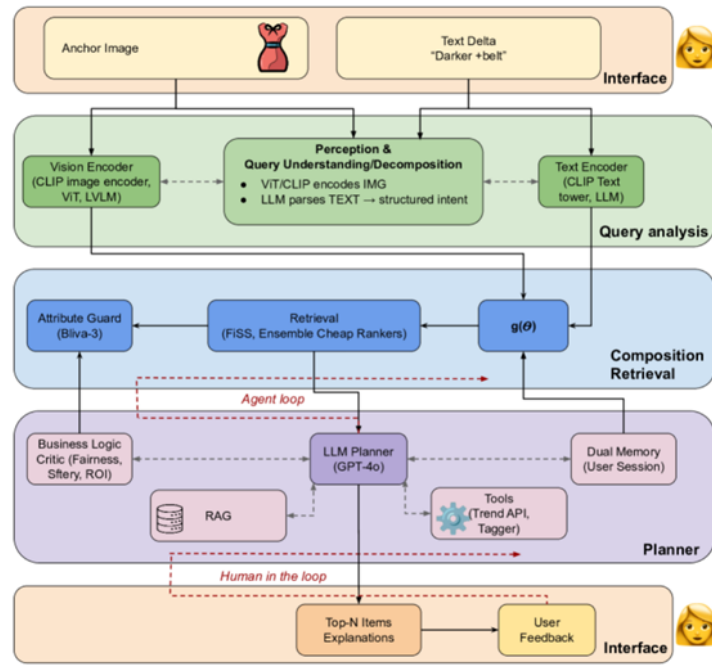
channel.” Although Li-Ning officially entered the outdoor market in 2024 and launched the “Wanlongjia” technical jacket series, finding a differentiated position in a market already overcrowded and highly segmented by Anta-affiliated brands remains its core commercial challenge [8]. At the end of 2025, Li-Ning opened its first independent outdoor concept store, Counterflow, in Beijing Chaoyang Joy City, attempting to transition the brand from “professional sports” to “urban outdoor lifestyle” through experimental retail scenarios [8]. This transition requires precise trend curation rather than mechanical data collection.

## 2. Existing Research and Technical Status

Current fashion trend analysis is undergoing an evolution from “expert sensation-driven” to “multimodal data intelligent curation” [12]. Traditional trend forecasting often relies on qualitative observations of fashion shows and street photography—a method that tends to have a lag effect when dealing with the rapidly changing digital business environment of 2026 [14], as summarized in Figure 4.

### 2.1 Application of Multimodal Large Models in the Fashion Field

Academia and industry have begun exploring the use of Vision-Language Models (VLM) to handle the complexity of fashion data. For example, the UniFashion model launched in 2024 uses a single framework to achieve cross-modal retrieval and generation, greatly improving the recognition accuracy of subtle clothing attributes (such as zippers, stitching, and fabric



**Figure 5.** Illustration of multimodal large-model applications in fashion understanding.

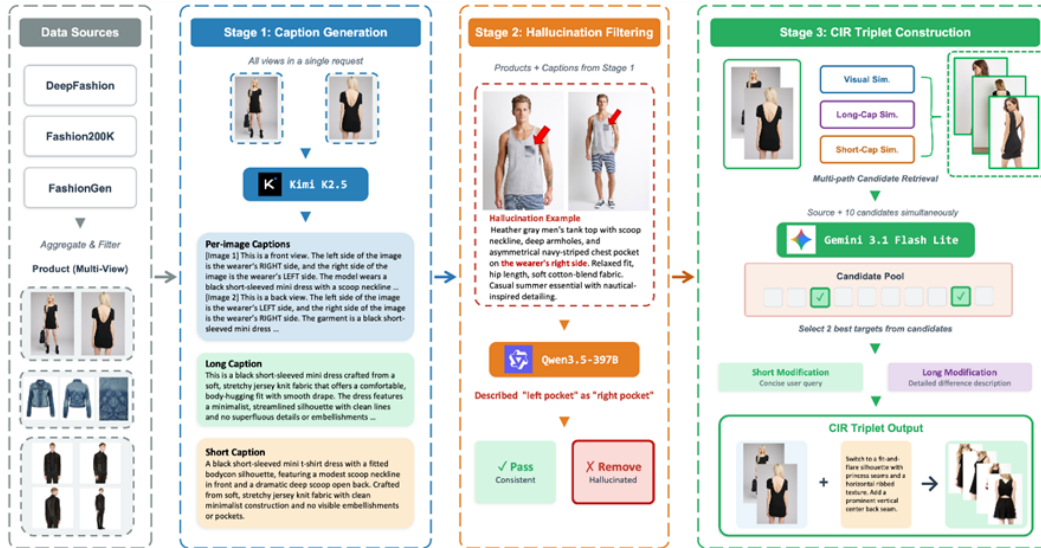
texture) [16]. Research shows that multimodal models such as Qwen-VL and GPT-4V can output structured JSON annotations for a single clothing image, covering fit types, material weights, and functional component layouts [14]. Furthermore, the Fashion-CLIP model, fine-tuned for the fashion domain, has become an industry standard, mapping images and text into the same vector space to capture subtle visual semantic deviations between “Urban Function” and “Hardcore Mountain Style” [18], as demonstrated in Figures 5 and 6.

## 2.2 Data Curation vs. Simple Scraping

Simple data scraping only yields fragmented information, whereas curation requires the construction of an interpretable and reusable knowledge base through cleaning, version control, metadata schema design, and ethical auditing [14]. In fashion recommendation systems, the introduction of agents is redefining the digital shopping journey. For instance, the AI assistant developed by Boyner provides “occasion-aware” matching suggestions based on real-time trend signals, providing a prototype reference for the design curation agent in this project [20].

## 3. Core Business Problem Identification

This project aims to develop a reusable AI data curation pipeline and a multi-agent system for leading Chinese sportswear brands (e.g., Anta and Li-Ning). By extracting color and



**Figure 6.** Workflow of cross-modal fashion analysis and structured annotation.

material pain points from unstructured data across platforms like Xiaohongshu, Bilibili, and e-commerce sites, the system translates these insights into actionable design recommendations for the “Lightweight Gorpcore” aesthetic.

### 3.1 Scenario Mismatch and Over-design

Hardcore outdoor gear typically utilizes 3L lamination technology for extreme waterproof performance. However, in the summer and transitional seasons of China’s first- and second-tier cities, such designs often lead to severe stuffiness and stiffness (“crumpling”) [1]. Feedback from bullet-chat data on Bilibili shows that “too heavy,” “not daily enough,” and “feels like wearing armor” are the top three reasons consumers abandon professional outdoor products [3]. Brands need a quantitative method to define the balance point of “lightweighting.”

### 3.2 Tension Between Brand Perception and Category Extension

Single-brand companies like Li-Ning often encounter the limits of “Brand Stretch” when expanding categories. When a brand represents basketball, badminton, and outdoor sports simultaneously, consumers find it difficult to develop specific functional trust [8]. 2025 annual reports show that while Li-Ning’s running business accounts for 31% of revenue, the sell-through conversion rate for the outdoor category remains much lower than that of Anta-affiliated brands [8]. Therefore, the goal of design curation is not just aesthetics but re-anchoring the brand’s professional credibility through visual language.



### 3.3 Data Silos and Curation Blind Spots

Current trend research often suffers from “Platform Bias.” Xiaohongshu data reflects “elite” and “filtered” trends, while real consumer pain points are often hidden in negative reviews on e-commerce platforms or anonymous bullet chats on Bilibili [7]. Designing based solely on the visual popularity of a single platform can lead to missing core functional requirements.

## 4. Research Objectives and Output Orientation

The ultimate goal of this project is to establish an AI-assisted Gorpcore design curation system for brands like Anta or Li-Ning.

1. **Build a Multimodal Gorpcore Meta-database:** Integrate data from Xiaohongshu (aesthetic side), Bilibili (pain point side), and e-commerce platforms (feedback side) to form a dataset containing at least 2,000 high-quality records with strict version control [14].
2. **Quantify the Evolution of Gorpcore:** Identify dynamic changes in design elements from 2023 to 2026. For example, analyze how pocket designs have shifted from pure capacity-driven to modular and “stealth” layouts [1].
3. **Develop “Gorpcore-Curator-Agent”:** Implement an automated curation pipeline using multimodal models to deeply parse images and align visual features with textual demands semantically [14].
4. **Provide Differentiated Design Curation Recommendations:** Offer specific suggestions for the next season’s product line based on analysis results, including optimal color matrices, material weight suggestions, and specific tailoring details (e.g., under-arm ventilation systems or multi-directional stretch seams) [1].
5. **Establish Risk and Ethical Guidelines:** Ensure the curation process complies with GDPR and China’s AI labeling regulations within the context of 2025’s strict privacy laws [21].

The overall framework of research objectives and output orientation is presented in Figure 7.

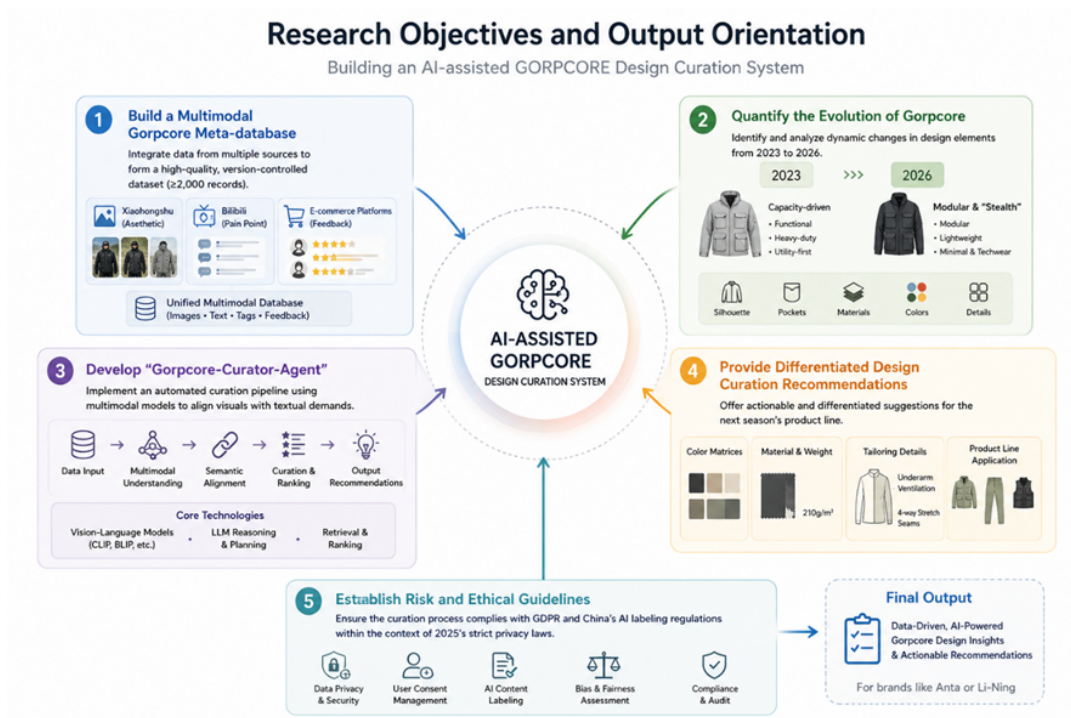


Figure 7. Framework of research objectives and overall output orientation.

## 5. Data Source Analysis and Curation Pipeline

This project adopts a “Trinity” multimodal data coverage strategy to achieve a closed loop between “visual perception” and “semantic feedback.”

### 5.1 Data Source Matrix

Our main data sources are depicted in Table. 1

### 5.2 Metadata Definition and Version Control

To ensure scientific curation, the dataset is divided into three phase versions:

- **Dataset v0 (Raw):** Unprocessed raw data containing 5,000+ records, mixed with scenery photos and advertisements [14].
- **Dataset v1 (Annotated):** Structured data annotated by the Multimodal Agent. Each record includes Visual\_Primary\_Color (HEX), Design\_Element (e.g., drawstrings, 3D pockets), Fit\_Type, etc. [14].
- **Dataset v2 (Verified):** Final curated library after semantic cross-verification. Noise points where visual descriptions conflict with semantic feedback are removed, and de-identification (e.g., face blurring) is completed [14].

Table 1. Data Source Matrix

| Platform              | Modality                                | Curation Value                                                                                                                                                  | Core Keywords                                                      |
|-----------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Xiao hong-shu         | High-res images + copy                  | Extracting visual aesthetic trends, color matrices, and matching logic. Reflects “idealized” Gorpcore demands.                                                  | #GorpcoreOutfits, #GorpcoreCom-mute, #MountainBoy                  |
| Bilibili              | Video frames + bullet chats + comments  | Capturing dynamic usage scenarios and immediate negative feedback (e.g., “too hot,” “not scratch-resistant”). Extracting technical parameters from pro reviews. | Outdoor Review, Arc’teryx Tear-down, Li-Ning Wanlongjia Comparison |
| E-commerce (Tmall/JD) | Structured re-views + negative feedback | Mining pain points from popular items of competitors (Arc’teryx, Salomon, Camel). Finding “niche markets” for product improvement.                              | Waterproof Failure, Zipper Jamming, Poor Fit                       |

5.2.1 Part I: Administrative & Provenance Metadata

Table 2. Administrative & Provenance Metadata Fields

| Field Name      | Description                | Specification/Controlled Vocabulary |
|-----------------|----------------------------|-------------------------------------|
| Source_ID       | Unique identification code | Format: GRP-2026-[Platform]-[Index] |
| Platform        | Scraping platform          | Xiaohongshu, Bilibili, Tmall, JD    |
| Timestamp       | Post/Crawl time            | ISO 8601 format                     |
| Privacy_Status  | Privacy status             | Pending, Face_Blurred, PII_Removed  |
| Dataset_Version | Data version               | v0 (Raw) or v1 (Curated)            |

### 5.2.2 Part II: Descriptive & Multimodal Metadata

**Table 3. Descriptive & Multimodal Metadata Fields**

| Field Name     | Description           | Extraction/Vocabulary                               |
|----------------|-----------------------|-----------------------------------------------------|
| Raw_Text       | Original copy/comment | String (processed by NLP)                           |
| Primary_HEX    | Primary visual color  | Extracted HEX code                                  |
| Secondary_HEX  | Accent color          | Used for color-blocking analysis                    |
| Visual_Objects | Identified components | Pockets, buckles, drawstrings, zippers (via YOLOv8) |
| Fit_Type       | Tailoring type        | Oversized, Regular, Tight, Cropped                  |

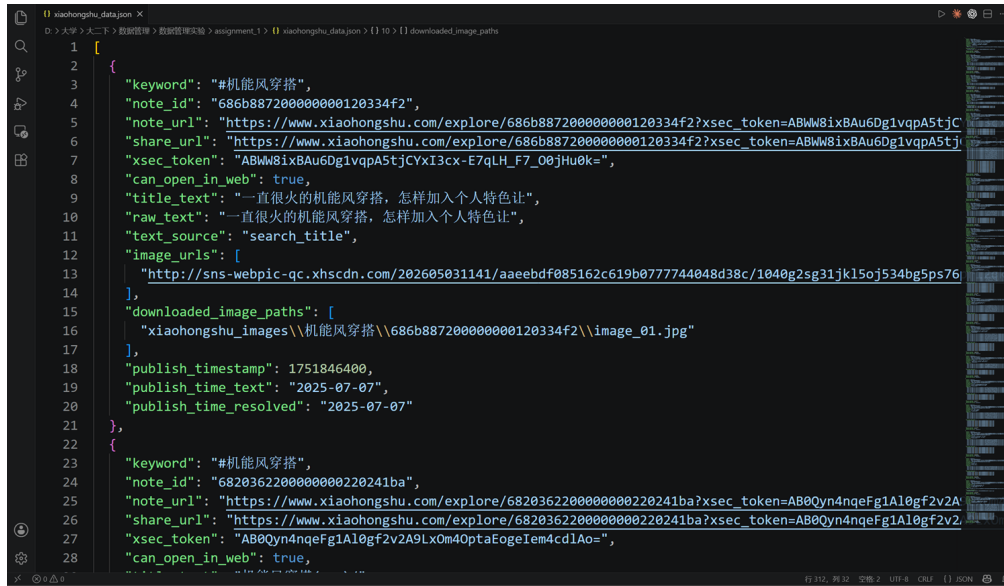
### 5.2.3 Part III: Analytical & Trend Intelligence Metadata

**Table 4. Analytical & Trend Intelligence Metadata Fields**

| Field Name      | Description      | Logic & Importance                                          |
|-----------------|------------------|-------------------------------------------------------------|
| Material_Clue   | Material tags    | Matte_Nylon, Gore-Tex, Reflective, Ripstop                  |
| Scenario        | Usage scenario   | Hardcore_Outdoor, Urban_Commute, Office                     |
| Pain_Point      | User pain points | From Bilibili (e.g., “Non-breathable,” “Overweight”)        |
| Signal_Strength | Signal strength  | Composite score (0–1) based on likes, shares, and sentiment |

## 5.3 Initial Raw Dataset Verification

To ensure the end-to-end technical feasibility of our pipeline, the team has successfully implemented an automated collection process, yielding the initial raw dataset designated as



**Figure 8.** Initial Raw Dataset V0 Verification (Partial)

Dataset v0.

**Data Composition and Multimodal Attributes:** As demonstrated in Fig. 8, the dataset captures unstructured raw inputs directly from social media platforms. Beyond basic platform metadata such as `note_id` and `publish_time`, the primary value lies in the established multimodal alignment:

- **Semantic Layer:** The preservation of `title_text` and `raw_text` provides the raw corpus for subsequent NLP tasks, including sentiment analysis and scenario identification.
- **Visual Layer:** The system not only logs the original cloud-based `image_urls` but also completes the localization mapping via `downloaded_image_paths`.

### Significance as a Curation Baseline:

Serving as Dataset v0, this collection represents the “Ground Zero” of our curation chain. It maintains full traceability, ensuring that every business insight can be audited back to its original social media context. The completion of this phase marks the transition from mere “Scraping” to professional “Curation.” By retaining these raw signals—including the inherent noise—we provide a high-fidelity environment for the Gorpcore-Curator-Agent to perform precise feature extraction and cross-modal verification in the upcoming stages.

## 6. Strategy and Research Path

The core competitiveness of this project lies in the construction of the “Gorpcore-Curator-Agent,” which transforms traditional manual curation logic into an extensible algorithmic process.

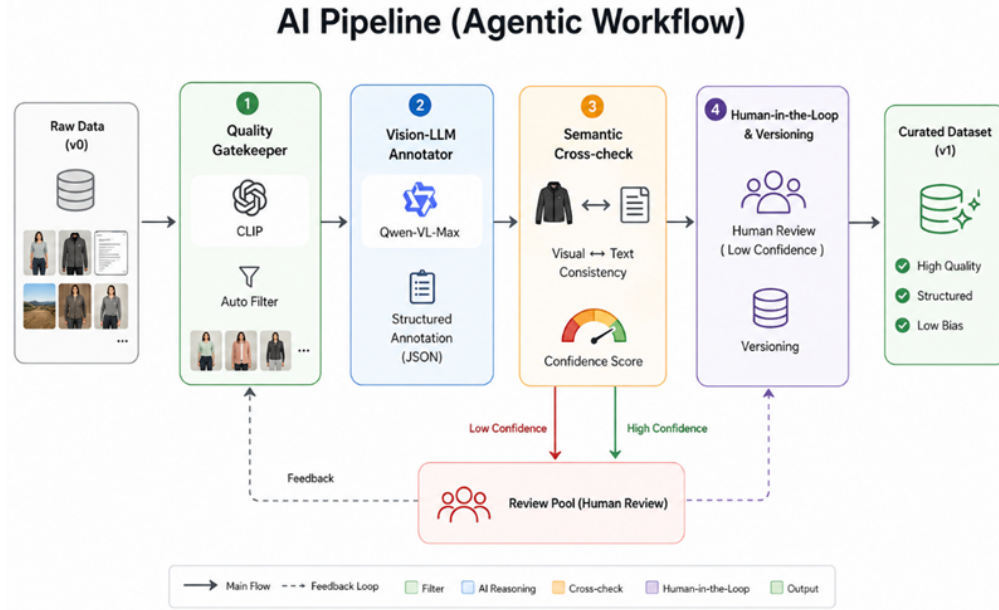
### 6.1 AI Pipeline (Agentic Workflow)

The agent is divided into three pipeline nodes with independent cognitive functions:

1. **Node A: Quality Gatekeeper:** Calls the CLIP model to determine if images in a million-level stream contain “full or half-body outfits.” Automatically filters out text screenshots, scenery, and repetitive marketing materials. This is the key to transitioning from “Scraping” to “Curation” [14].
2. **Node B: Vision-LLM Annotator:** Employs the **Qwen-VL-Max** multimodal large model. The agent receives images and performs reasoning based on structured prompts. Example Prompt: “Analyze this image for Gorpcore elements. Output JSON: 1. Pocket count. 2. Zipper type (Sealed/Exposed). 3. Fit (Loose/Tight). 4. Visibility elements (Reflective strips).” [14].
3. **Node C: Semantic Cross-check:** A core step to correct data bias. The agent retrieves associated text. If the copy claims “suitable for summer commute” but visual features show “heavyweight composite fleece,” the confidence score is lowered. This identifies “fake grass-planting” signals on social media [12].
4. **Node D: Human-in-the-Loop & Versioning:** Automation does not replace human judgment. Data with low confidence scores enter a review pool for team members to provide final qualitative assessments based on subcultural context. After this four-step pipeline, raw data (v0) is condensed into a high-quality curated dataset (v1), as illustrated in Figure 9.

### 6.2 Visual Innovation: Semantic Feature Visual Word Cloud

This project proposes a “Visual Word Cloud” method. Using **YOLOv8**, the system locates and crops specific design components (e.g., 2,000 different cuff velcros or pocket flaps). Utilizing semantic vectors generated by **DINOv2** or **Fashion-CLIP**, it performs dimensionality reduction clustering (t-SNE) to generate a trend concept map composed of real physical components [14].



**Figure 9.** Workflow of the Gorpcore-Curator-Agent.

**Business Significance:** This method bridges the communication gap between “data analysts” and “fashion designers.” It outputs a visualization mood board driven by real consumer data (Data-driven Mood Board). Designers can intuitively perceive the optimal selection of accessories, pocket angles, and drawstring preferences.

## 7. Technical Infrastructure and System Architecture

### 7.1 Tech Stack Configuration

- **Data Acquisition:** Playwright-based automation cluster to capture Xiaohongshu waterfall stream data [14].
- **Computing & Inference:**
  - Visual Semantic Encoding: **OpenCLIP-ViT-L-14**.
  - Object Detection: **Ultralytics YOLOv11** (latest version optimized for small component recognition) [14].
  - LLM Driven: **LangGraph** framework to coordinate multi-agent workflows.
- **Visualization:** Streamlit-based interactive dashboards; report typeset in **L<sup>A</sup>T<sub>E</sub>X** [14].

### 7.2 System Architecture Logic

The system is divided into five logical levels:

- **Input Layer:** Integrates social media, video bullet chats, and e-commerce competitor reviews.
- **Governance Layer:** Executes data deduplication, face blurring, and privacy field discarding.
- **Parsing Layer:** Agents execute multimodal feature extraction.
- **Analytics Layer:** Executes correlation analysis (e.g., HEX values vs. sales conversion).
- **Synthesis Layer:** Transforms results into brand decision reports and visual word clouds.

The complete system architecture is illustrated in Figure 10.

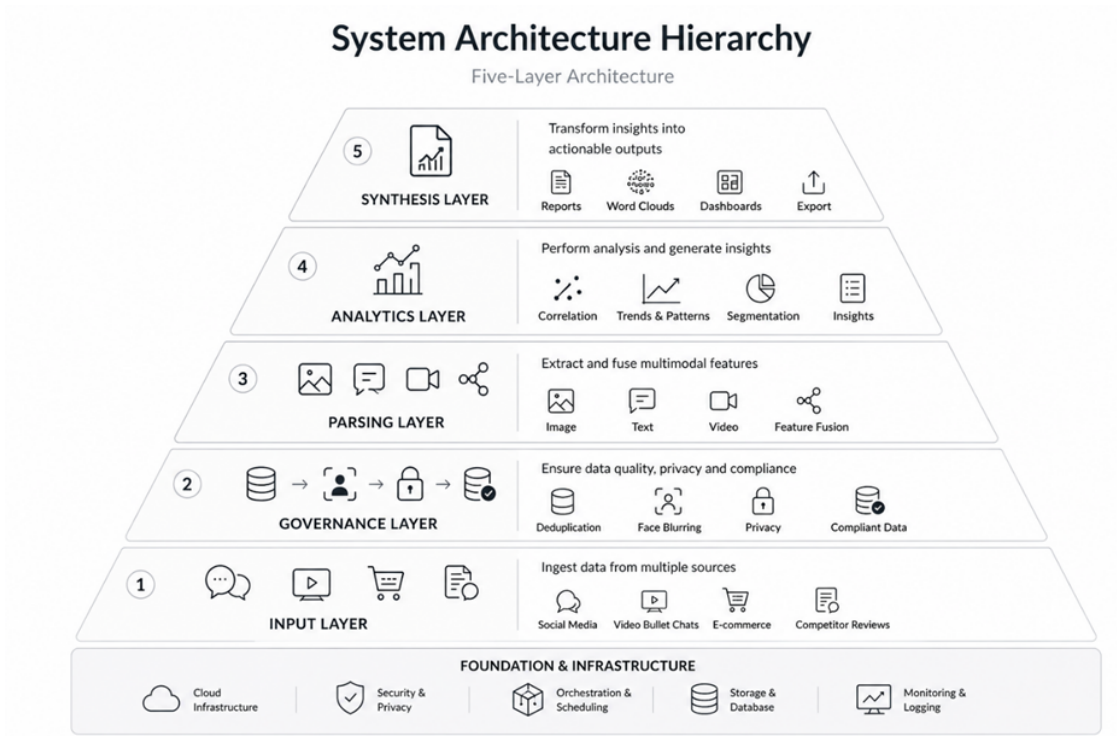


Figure 10. Diagram of the technical system architecture and logical layers.

8. Project Roadmap and Milestones

This project adopts an agile development model, divided into four key implementation phases based on the functional modules, as summarized in Table 5.



Table 5. Project Module Overview

| Module Name                               | Core Work Content                                                                                                                                                                                                                          | Core Deliverables                                                                    |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Data Acquisition                          | Build automated crawling scripts based on Playwright/Selenium to capture Xiaohongshu outfit graphics; obtain video bullet screens and comments through Bilibili API; crawl user reviews of hot-selling products of competitors on Tmall/JD | Dataset v0 (Raw Dataset),<br>Raw Data Metadata List                                  |
| Data Cleaning and Privacy Desensitization | Deduplication and noise filtering of raw data; text data cleaning and keyword extraction; face desensitization and compliance processing of image data; dataset standardization                                                            | Dataset v1 (Curated and Compliant Dataset), Data Cleaning and Desensitization Report |
| Multimodal Feature Extraction             | Extract image color features based on OpenCV; realize cross-modal vector mapping of graphics and text based on CLIP model; realize structured annotation of design elements based on multimodal large models                               | Structured Feature Database, Cross-modal Clustering Analysis Results                 |
| AI Automated Curation                     | Build the “Gorpcore-Curator-Agent” to complete data quality verification, automatic multi-dimensional feature annotation, and semantic cross-validation; generate semantic feature visual word cloud                                       | Automated Curation Report, Trend Visual Concept Map                                  |

Continued on next page

(Table 5 continued)

| Module Name                    | Core Work Content                                                                                                                                                                              | Core Deliverables                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Design Strategy Transformation | Integrate feature analysis results and user pain points, output full-dimensional design strategies for color, tailoring, materials, and accessories, and complete brand adaptation calibration | Lightweight Gorpcore Product Design Implementation Plan for Target Brand’s Next Season |
| Report Delivery                | Complete the full academic report type-set in LaTeX, including ethical statement, bias control, method description, conclusions and recommendations                                            | Final Academic Report, Full Set of Design Curation Exhibition Materials                |

8.1 Phase 1: Data Acquisition and Preprocessing (Week 1–2)

This phase integrates **Module 1** and **Module 2** to establish the data foundation.

- **Work Content:** Build automated crawling scripts based on Playwright/Selenium to capture Xiaohongshu outfit graphics; obtain video bullet screens and comments through Bilibili API; crawl user reviews of hot-selling products from competitors on Tmall/JD. This is followed by deduplication, noise filtering, text data cleaning, face desensitization, and dataset standardization.
- **Deliverables:** Dataset v0 (Raw Dataset), Raw Data Metadata List, and Dataset v1 (Curated and Compliant Dataset).

8.2 Phase 2: Multimodal Feature Engineering (Week 3–5)

Focusing on **Module 3**, this phase performs deep data annotation and mapping.

- **Work Content:** Extract image color features based on OpenCV; realize cross-modal vector mapping of graphics and text based on the CLIP model; realize structured annotation of design elements using multimodal large models.
- **Deliverables:** Structured Feature Database and Cross-modal Clustering Analysis Results.

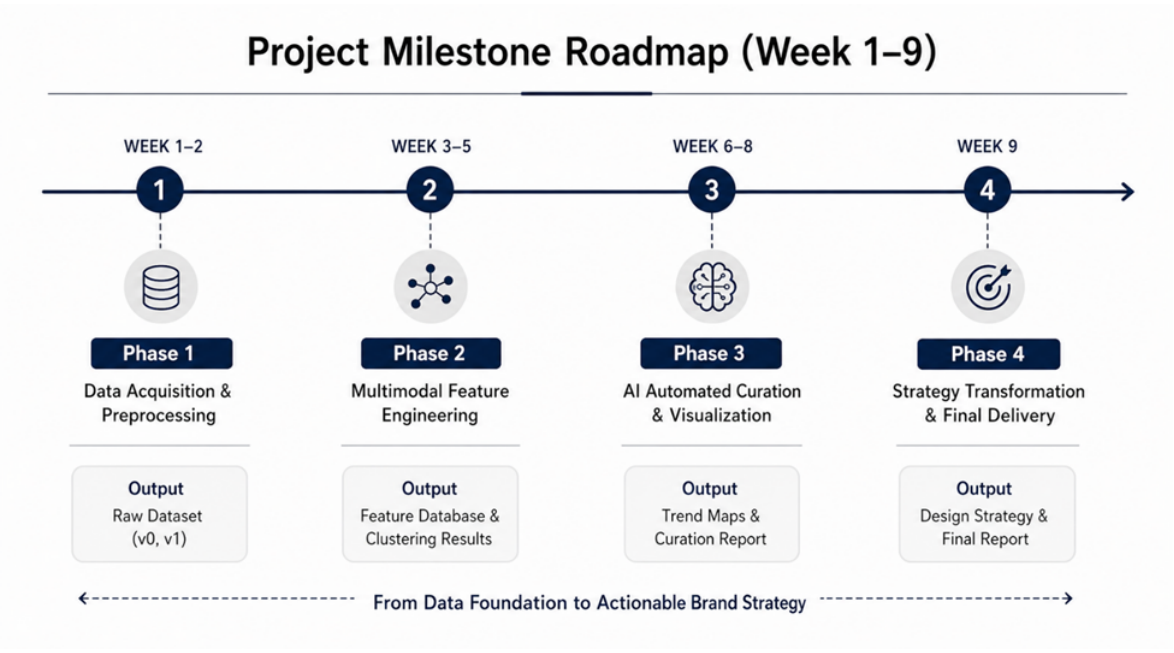


Figure 11. Project roadmap and phased milestone diagram.

8.3 Phase 3: AI Automated Curation and Visualization (Week 6–8)

Execution of **Module 4** to transform data into curated design insights.

- **Work Content:** Build the “Gorpcore-Curator-Agent” to complete data quality verification, automatic multi-dimensional feature annotation, and semantic cross-validation; generate semantic feature visual word clouds as trend concept maps.
- **Deliverables:** Automated Curation Report and Trend Visual Concept Map.

8.4 Phase 4: Strategy Transformation and Final Delivery (Week 9)

Finalization through **Module 5** and **Module 6** to output actionable brand strategies.

- **Work Content:** Integrate feature analysis results and user pain points to output full-dimensional design strategies for color, tailoring, materials, and accessories; complete brand adaptation calibration. Finalize the academic report typeset in LaTeX, including ethical statements and method descriptions.
- **Deliverables:** Lightweight Gorpcore Product Design Implementation Plan for target brand’s next season, Final Academic Report, and full set of Design Curation Exhibition Materials.

The full project roadmap and milestone schedule are shown in Figure 11.

9. Risk Control and Mitigation

Table 6. Risk Control and Mitigation Strategies

| Risk Item               | Potential Impact                                                | Mitigation Strategy                                                                            |
|-------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Data Compliance         | Regulatory review of AI training data in 2025.                  | Strict adherence to robots.txt; collect non-PII fields only [21].                              |
| Filter Bubble           | Agent only captures specific styles due to platform algorithms. | “Keyword randomization” and “multi-account” strategies; include global Fashion Week data [13]. |
| Technical Hallucination | VLM misidentifying fabric (e.g., DWR coating).                  | 10% random sampling for manual review by textile experts [12].                                 |
| System Latency          | VLM API responsiveness delays progress.                         | Offline asynchronous batch processing during low-traffic periods (late night).                 |

10. Ethical Review and Legal Framework

10.1 Privacy & Compliance

- **Face Blurring:** MTCNN classifier detects faces and applies 20×20 Gaussian blurring [21].
- **PII Removal:** No usernames, unique IDs, or specific geolocation kept in Dataset v2.
- **AI Labeling:** All AI-generated designs are labeled “AI-Generated” as per 2025 regulations [21].

10.2 Legal Boundaries

Follows the spirit of *hiQ Labs vs. LinkedIn*. Only crawls publicly indexed data without requiring login [21].



## **12. Conclusion**

The evolution from “Hardcore Outdoor” to “Urban Commute” epitomizes the transformation of China’s sports industry from extensive expansion to refined operation. Success for Anta and Li-Ning will depend on who can more accurately capture the real urban demand for “comfort, decency, and security” through data curation [4]. This system establishes a technical standard for the digital transformation of Chinese fashion brands within a framework of legal compliance and ethical governance.

Gorpcore’s essence is not clothing, but the peace and control humans attempt to reshape in the complex urban jungle of 2026 through fabrics and algorithms [3].

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